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Table 9.4: Anatomical differences between dicot stem and monocot stem			
S.No.	**Characters**	**Dicot Stem**	**Monocot Stem**
1.	Hypodermis	Collenchymatous	Sclerenchymatous
2.	Ground tissue	Differentiated into cortex, endodermis and pericycle and pith	Not differentiated, but it is a continuous mass of parenchyma.
3.	Starch Sheath	Present	Absent
4.	Medullary rays	Present	Absent
5.	Vascular bundles	(a) Collateral and open	(a) Collateral and closed
		(b) Arranged in a ring	(b) Scattered in ground tissue
		(c) Secondary growth occurs	(c) Secondary growth usually does not occur.

Table 9.5: Anatomical differences between root and stem			
S.No.	**Characters**	**Root**	**Stem**
1.	Epidermis	Absence of cuticle and epidermal pores.	Presence of cuticle and epidermal pores.
		Presence of unicellular root hairs.	Presence of unicellular and multicellular trichomes
2.	Outer Cortical cells	Chlorenchyma absent	Chlorenchyma present
3.	Endodermis	Well defined	ill-defined or absent.
4.	Vascular bundles	Radial arrangement	Conjoint arrangement
5.	Xylem	Exarch	Endarch

Anatomy of a Dicot and Monocot Leaves

Leaves are very important vegetative organs. They are mainly concerned with **photosynthesis and transpiration**. Like stem and roots, leaves also have the three tissue system – dermal, ground and vascular. The dermal tissue system consists of an upper and lower epidermis. The ground tissue system that lies between the epidermal layers of leaf is known as **mesophyll tissue**. Often it is differentiated into **palisade parenchyma** on the adaxial (upper) side and **spongy parenchyma** on the abaxial (lower) side.

In dorsiventral leaves the mesophyll is differentiated into palisade and spongy parenchyma, the former occurring on the upper side and the later on the lower side Example: Sunflower. In isobilateral leaf palisade is present on both sides of the leaf and inbetween them spongy parenchyma is present. Example: Nerium. In some plants Example: Ficus calcium crystals are present. There are also leaves where spongy tissue alone is present in some epidermal cells Example: Grasses.

The presence of air spaces is a special feature of spongy cells. They facilitate the